Regression Algorithms for Learning & Its Application on Crypto Currency Markets

Project Plan

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Undergraduate Final Year Project

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1. Abstract

Regression Algorithms are a core part of supervised machine learning. They provide methods for modelling relationships between multiple variables and predicting continuous outcomes. A specific type of Regression Algorithm known as Ridge Regression is useful in addressing issues of multicollinearity and overfitting. Ridge Regression is a variant of linear regression that introduces a L2 regularization term (Hastie, Tibshirani & Friedman, 2009).

This project aims to implement ridge regression and apply it to a real-world financial dataset. To be specific the project will be focusing on the BTC-USD market. The objective of this project is to analyse and evaluate the ability of ridge regression in being able to predict the short-term price action of cryptocurrency. Furthermore, the project will compare this performance with other regression models such as standard linear regression and possibly Lasso or KNN regression.

The cryptocurrency market provides an ideal platform to test regression algorithms due to its volatility, continuous trading and the high correlation between the price movement and the trading volume. The data used in the project will be acquired from reliable financial APIs, feature engineering like Moving Averages and Rolling Volatility, as well as the implementation of ridge regression using Python's scikit-learn library (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, & Duchesnay, 2011).

In the first term, the project will aim to develop foundational reports and proof-of-concept programs portraying theoretical understanding and algorithmic correctness. In the second term, the project will build on and progress towards complete object-oriented software implementation, featuring a user interface and visual analysis of performance metrics.

All in all, the project looks to exemplify how ridge regression can be used on highly volatile financial data to assess its robustness, interpretability, and generalization capabilities in order to contribute to the comprehension of regression-based predictive modelling in the financial world.

2. Timeline

As stated previously, term 1 will focus on foundational reports and proof-of-concept while term 2 will focus on object-oriented implementation.

Term 1

Week 3-4	Study the mathematical foundations of regression algorithms, focusing on linear, ridge, and lasso regression (Hastie, Tibshirani, & Friedman ,2009; Bishop, 2006)
Week 5-6	Write the first report: an overview of regression algorithms, explaining the training/test set concept, cost functions, and derivation of the ridge regression formula. Implement a proof-of-concept program using a small synthetic dataset (via NumPy).
Week 7-8	Conduct a literature review on financial forecasting using regression models (Fischer & Krauss, 2018; Patel, Shah, Thakkar, Kotecha, 2015). Write the second report summarizing prior applications of regression models to financial and cryptocurrency data. Report 3 on Examples of applications of Ridge Regression worked out on paper and checked using the prototype program.
Week 9-10	Load and preprocess real-world Bitcoin price data. Perform exploratory data analysis, correlation studies, and feature normalization. Implement ridge regression on preprocessed BTC-USD data and evaluate using MSE and R^2 metrics. Write the third report presenting preliminary results and model performance.
Week 11	Finalize the first-term deliverables and prepare for the interim presentation.

Term 2

Winter Break & Week 1-2	Implement a full Object-Oriented structure to the Ridge Regression implementation using Python
Week 3-4	Develop a GUI using Tkinter or PyQt to allow user interaction and visualization of regression results.
Week 5-6	Implement and compare a second regression model such as KNN or Lasso Regression.
Week 7-8	Conduct experiments comparing models, analyzing overfitting, regularization strength, and stability.
Week 9-10	Write the final project report describing theoretical foundations, software engineering process, experimental results, and discussion. Prepare a presentation for it.

3. Risks and Mitigations

Risks are part of every project and this project is no exception. Most risks can be mitigated using a counteractive measure however, some are unavoidable. In this section, some risks and their possible mitigation strategies will be discussed.

3.1 Data Quality and Availability

Risk: Due to the nature of Cryptocurrency markets there is a risk that the data can include missing values, anomalies, or inconsistencies.

Mitigation: To mitigate this, the project will use verified APIs such as Yahoo Finance or Binance and clean datasets through interpolation and smoothing. A local backup of processed data will be kept just in case.

3.2 Overfitting and Model Misinterpretation

Risk: The model might perform well with the training data but generalize poorly with unseen data.

Mitigation: To mitigate this, the project will use cross-validation, regularization tuning, and carry out test sets to ensure fair evaluation.

3.3 Computational Efficiency

Risk: Processing large time-series data and performing parameter optimization may be time-consuming which induces the risk of inefficiency.

Mitigation: To mitigate this, the project will use efficient libraries (NumPy, Pandas, scikit-learn) and limit parameter searches through grid or random search strategies.

3.4 Time Management

Risk: There is a risk that the imbalance between programming and report writing could lead to time delays affecting the project's timeline.

Mitigation: To mitigate this, the project will adhere to the strict weekly plan outlined in the previous section. There will also be intermediate deadlines and Gitlab will be used for progress tracking and version control.

3.5 Evaluation Bias

Risk: Relying on a single performance metric may lead to the risk of misrepresentation of model capabilities.

Mitigation: To mitigate this, the project will use multiple metrics such as MSE, MAE, R^2, and directional accuracy for a balanced and more complete evaluation.

3.6 Dependency Conflicts

Risk: Library incompatibilities or API changes disrupting data collection is another risk that the project can encounter.

Mitigation: To mitigate this, the project may use virtual environments. It will also document dependencies in a requirements.txt file, and cache data locally.

3.7 Data Bias and Non-Stationarity

Risk: Financial and Cryptocurrency data are often non-stationary meaning their patterns and distributions change over time. This could lead to a risk where the model is trained on old data and performs poorly with new data.

Mitigation: To mitigate this, the project could use a strategy like rolling-window which retrains the model periodically with more recent data. The project could also include tests to ensure model validity over time.

3.8 Ethical and Responsible AI Considerations

Risk: The use of machine learning models for financial predicting could lead to ethical misuse if interpreted as financial advice.

Mitigation: To mitigate this, the project will clearly state that the model is developed for academic and research purpose only and should not be considered as financial advice for investment or trading decisions and that the project holds no liability in such a case.

3.9 Backup and Version Control

Risk: Loss of code or data due to device failure or accidental deletion is another risk that the project needs to account for.

Mitigation: To mitigate this, the project will store all code and documents on Gitlab with regular commits and backups to ensure that the code or data is not lost.

4. Acronyms and Glossary

Acronyms

1. BTC Bitcoin
2. USD United States Dollar
3. KNN K-Nearest Neighbors
4. API Application Programming Interface
5. MSE Mean Squared Error
6. R^2 Coefficient of Determination
7. GUI Graphical User Interface
8. MAE Mean Absolute Error

Glossary

1. Regression: A statistical method used to model the relationship between a dependent variable (target) and one or more independent variables (features).
2. Ridge Regression: A linear regression method that includes a L2 regularization term to penalize large coefficients and reduces overfitting.
3. Regularization: A method to restrict model complexity for improved generalization.
4. L2 Regularization: A penalty used in Ridge Regression that minimizes the sum of squared coefficients.
5. Cross-Validation: A method for assessing how well a model generalizes to unseen data.
6. Feature Engineering: The process of transforming raw data into useful features that enhance model performance.
7. Overfitting: When a model excessively fitted to the training data instead of the underlying data pattern and performs poorly on unseen data.
8. Test Set: The part of data used to evaluate the performance of the trained model.
9. Model Generalization: The ability of a machine learning model to perform well on unseen data.
10. MSE: A metric that measures the average squared difference between predicted and actual values.
11. MAE: A metric that measures the average of absolute differences between predicted and true values.
12. R^2 Score: A measure that explains how much of the target's variance is captured by the model.

5. Bibliography

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